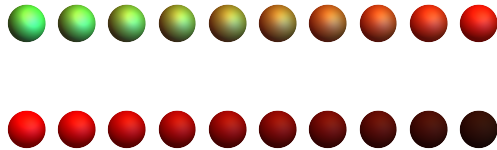


The Game of Life



When I was 15 I got interested in John Conway's Game of Life reading an article about it in the October 1970 issue of the *Scientific American* magazine.

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Computers, free will, and your mind

The Game of Life has figured in a lot of arguments about free will.

Many people argue the Game of Life has free will because *in principle* it is unpredictable. But being unpredictable means the patterns must also be unpredictable *to themselves*, which isn't what I take free will to mean.

Although a single brain synapse or neuron can't do much more than a cell in the Game of Life, your amazing brain has around 1,000 trillion (1,000,000,000,000,000) synapses all working together, ending up with the miraculous complexity of all your memories, your thinking, and your appreciation of science and art.

With only 150 times more cells than than this exhibit, the Game of Life can simulate a fully-working programmable computer.

It's a challenging thought that if the Game of Life was scaled up to millions of cells it would be capable of artificial intelligence, and it could be made so tiny many would fit on a pin head.

One of my favourite examples of how surprising the Game of Life is is that the Game of Life can play the Game of Life itself.

That's a bit like what your own consciousness does when you think about yourself. So, does that mean the Game of Life can be conscious too?

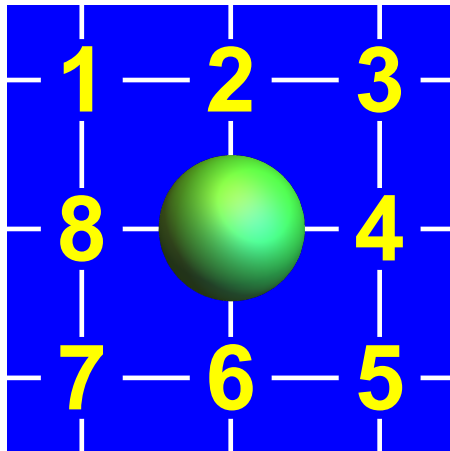
There are no scientific reasons why not, but its consciousness would be very different from ours.

Download links and the open source Python program I wrote to run the Game of Life at:
<https://www.harold.thimbleby.net/life>

Very simple rules

This small exhibit is enough to show how complexity can emerge from simple rules.

Each cell in the Game of Life is surrounded by up to eight other neighbouring cells, which may be dead or alive.



In the exhibit here, newborn cells are green, and as they age they are turned red then brown then black, like autumn leaves.

Just four rules

1 Under-population

Any live cell with fewer than two live neighbours dies.

2 Survival

Any live cell with two or three live neighbours lives on to the next generation.

3 Reproduction

Three live neighbours reproduce to create a new live cell.

4 Over-population

Any live cell with more than three live neighbours dies.

That's all there is to it.

In the exhibit here the rules are run every fraction of a second to create a new generation, using a Raspberry Pi computer behind the scenes.

If you watch the exhibit, you can see how the rules work.

The Game of Life is so popular people spend their time looking for interesting patterns. Bill Gosper discovered the first "glider gun" in 1970 which fires off lots of "gliders." The exhibit here shows a glider gun, some gliders, and a few other popular life forms.

The traffic lights (bottom left of the pre-planned show, and often found somewhere on the random show) are a good place to appreciate how the simple rules can create action.

Email me, or see the Wikipedia article *Conway's Game of Life* for lots more info.